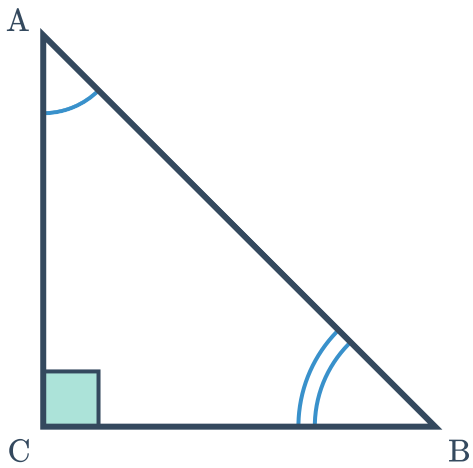
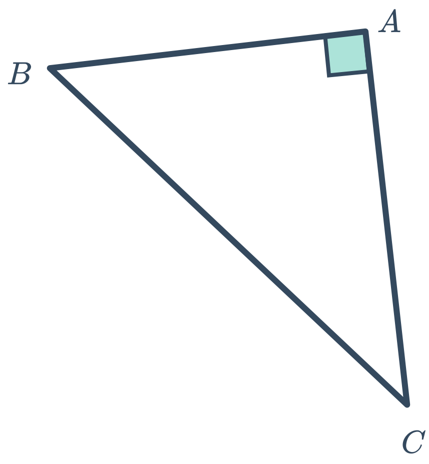
Right-angled triangles

1 For the following triangles, name the hypotenuse:

a



b



2 Consider the following triangle:

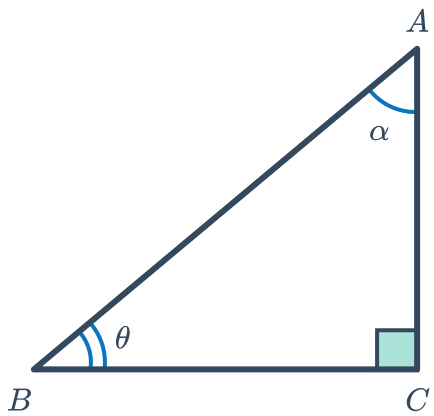
a State the opposite side to angle θ.

b State the adjacent side to angle θ.

c State the opposite side to angle α.

d State the adjacent side to angle α.

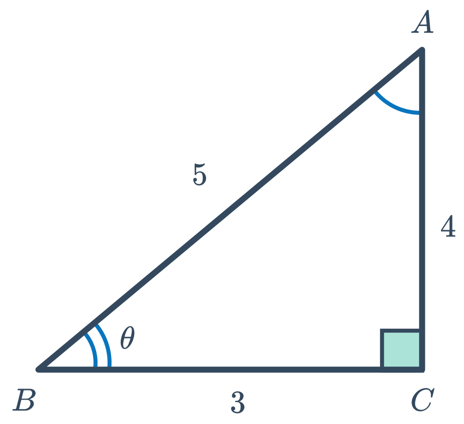
e State the angle that is opposite the hypotenuse.



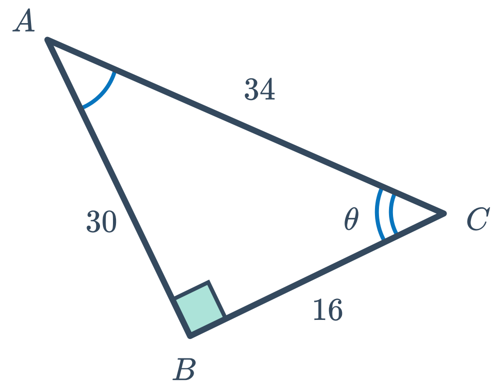
3 With reference the angle θ, find the value [redacted]se ratios for each of the following triangles:

i $\dfrac{\text{Opposite}}{\text{Adjacent}}$ **ii** $\dfrac{\text{Opposite}}{\text{Hypotenuse}}$ **iii** $\dfrac{\text{Adjacent}}{\text{Hypotenuse}}$

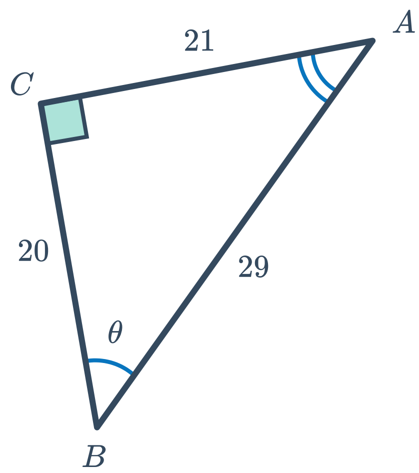
a



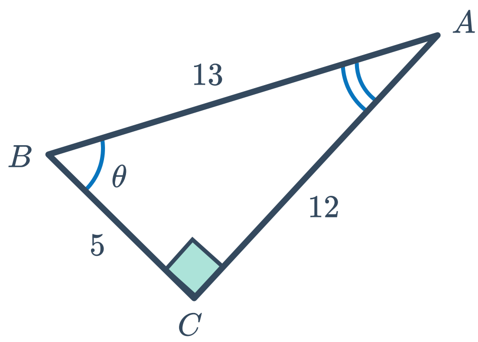
b



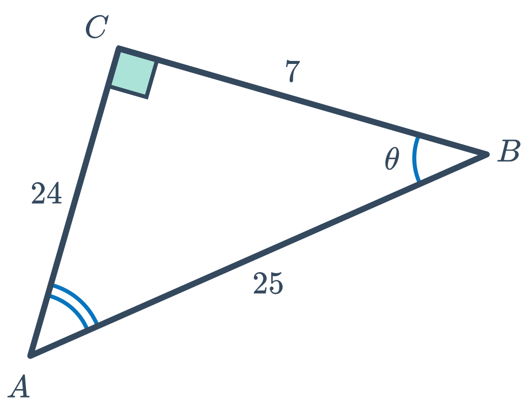
c



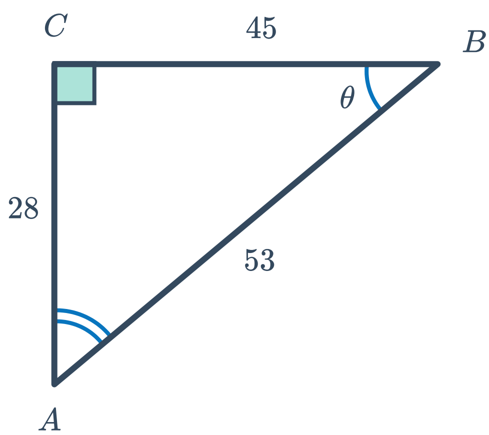
d



e



f

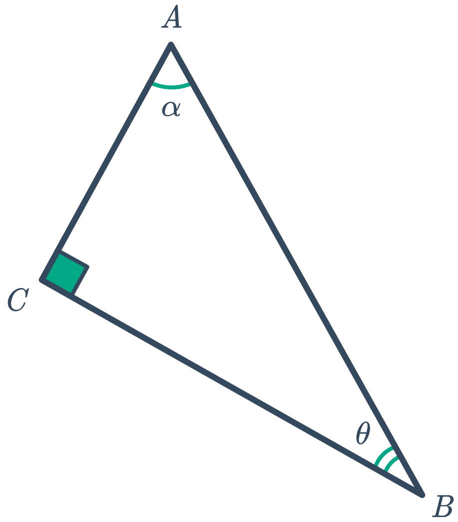


Trigonometric ratios

4 Write down the indicated ratios for the following triangles:

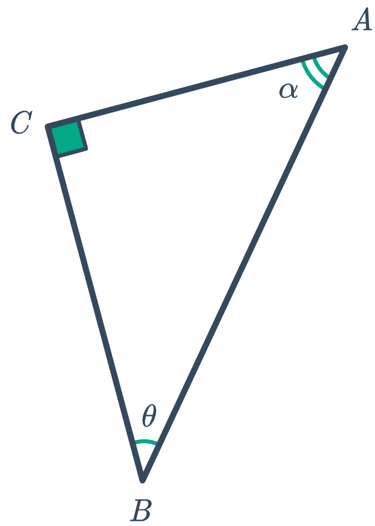
a

i $\tan\theta$ **ii** $\sin\alpha$



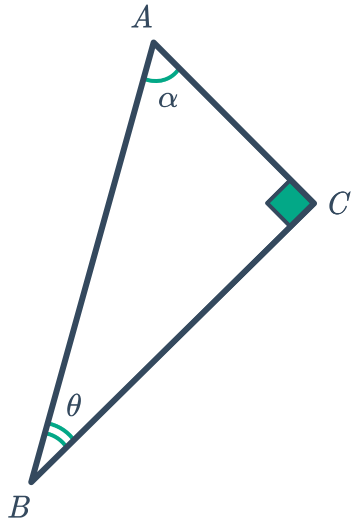
b

i $\sin\theta$ **ii** $\tan\alpha$



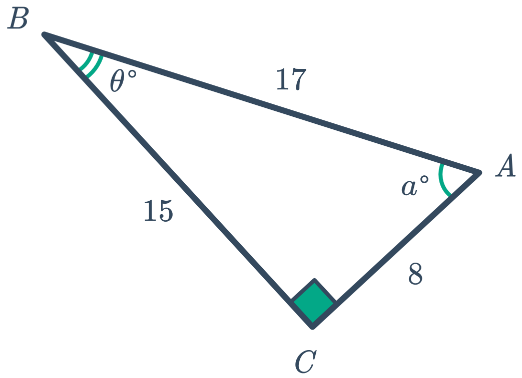
c

i $\cos\theta$ **ii** $\cos\alpha$



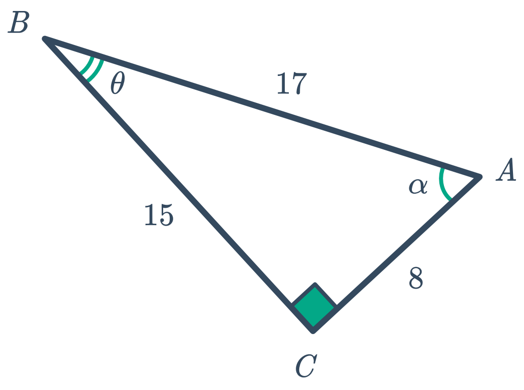
d

i $\tan\theta$ **ii** $\sin\alpha$



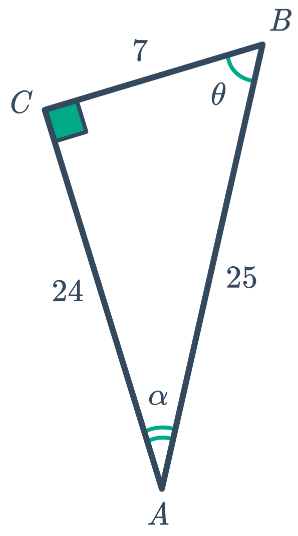
e

i $\cos\theta$ **ii** $\tan\alpha$



f

i $\sin\theta$ **ii** $\cos\alpha$

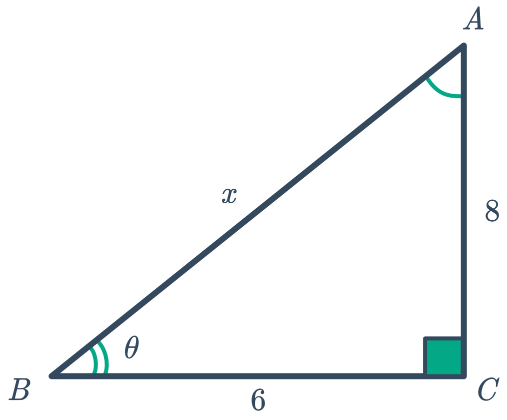


5 Consider the following triangle:

a Find the value of x.

b Find the value of $\sin\theta$.

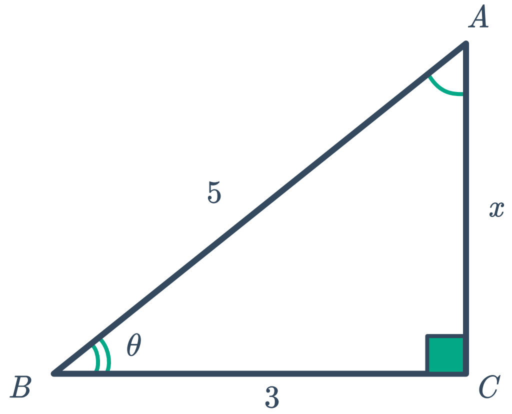
c Find the value of $\cos\theta$.



6 Consider the following triangle:

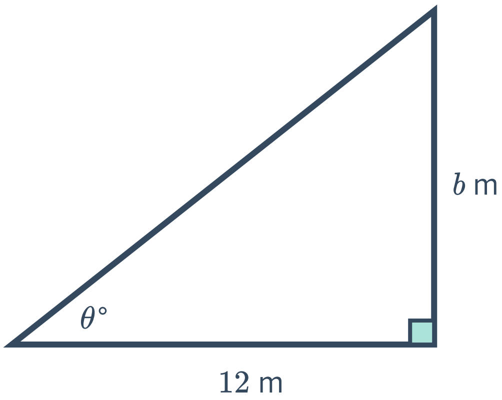
a Find the value of x.

b Hence, find the value of $\tan\theta$.

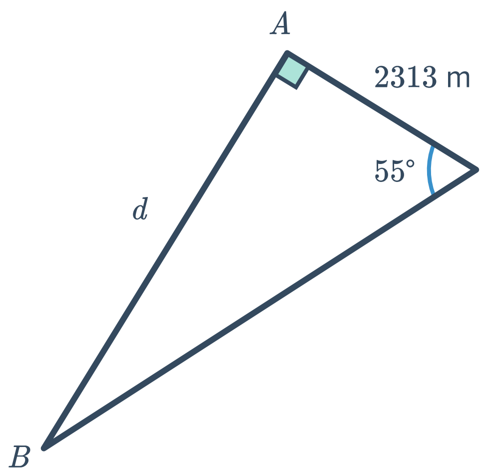


Unknown sides

7 For the following triangle, if $\tan\theta = \frac{2}{3}$, find the value of b.

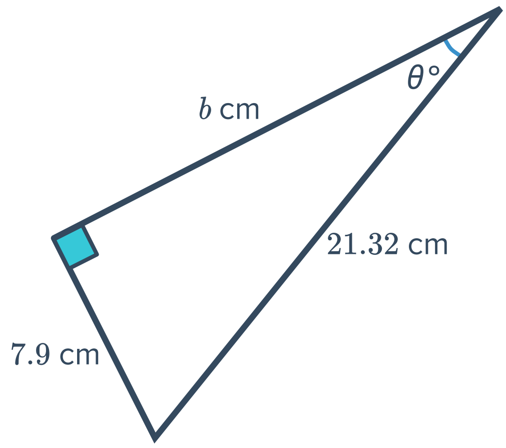


8 For the following triangle, if $\tan\theta = \frac{4}{3}$, find the value of d.



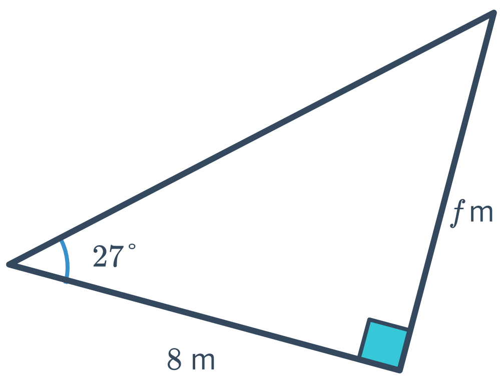
9 For the following triangle, if $\tan\theta = 0.4$, fi[redacted] the value of b.

Round your answer correct to one decimal place.

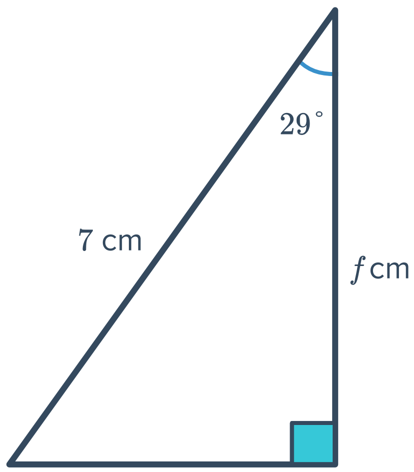


10 Find the value of the pronumeral in the following triangles, correct to two decimal places:

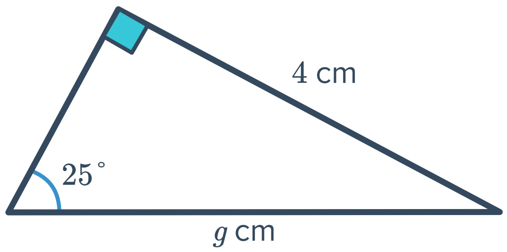
a



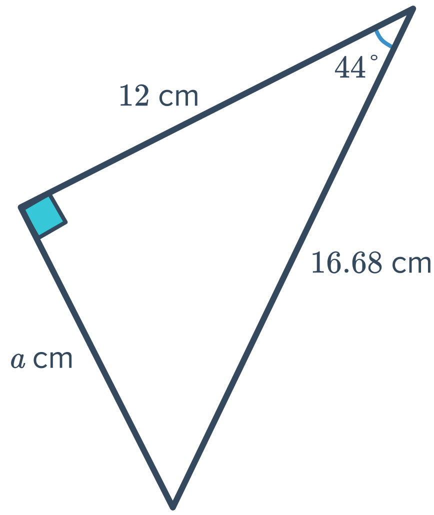
b



c

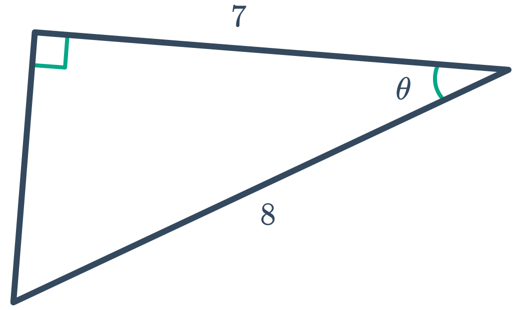


d



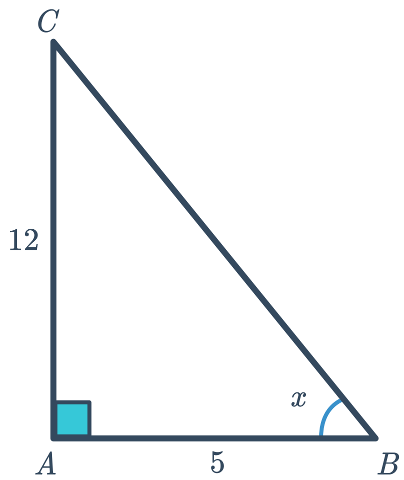
Unknown angles

11 Given the following triangle, calculate the exact value of $\tan\theta$.

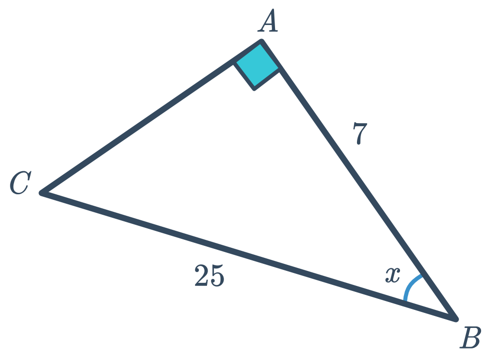


12 For each of the following triangles, find the value of x to the nearest degree:

a

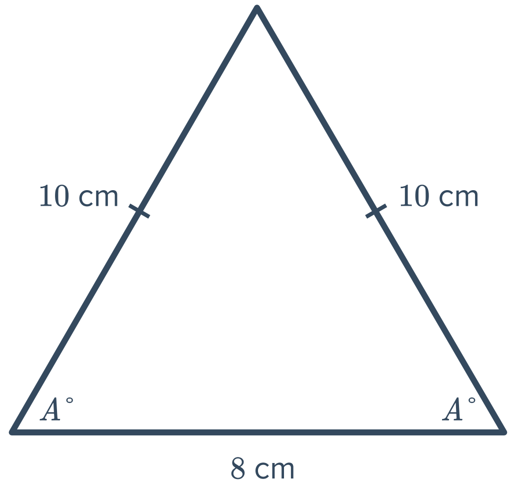


b

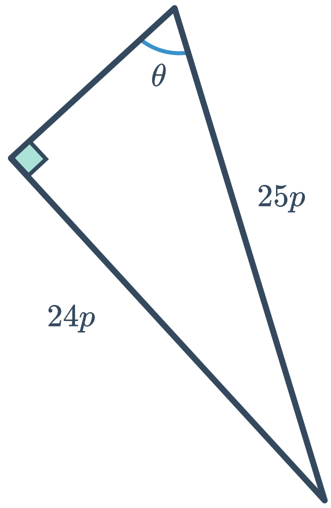


13 An isosceles triangle has equal side length of $10\ \mathrm{cm}$ and a base of $8\ \mathrm{cm}$ as shown.

Calculate the size of angle A to one decimal place.



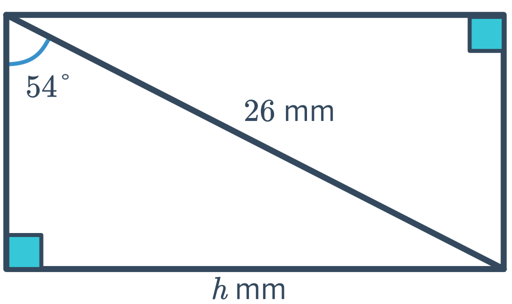
14 Find the value of $\tan\theta$ for the following triangle:



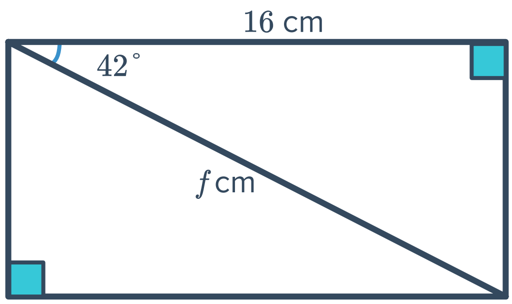
Applications

15 Find the value of the pronumeral(s) in the following diagrams, correct to the nearest whole number:

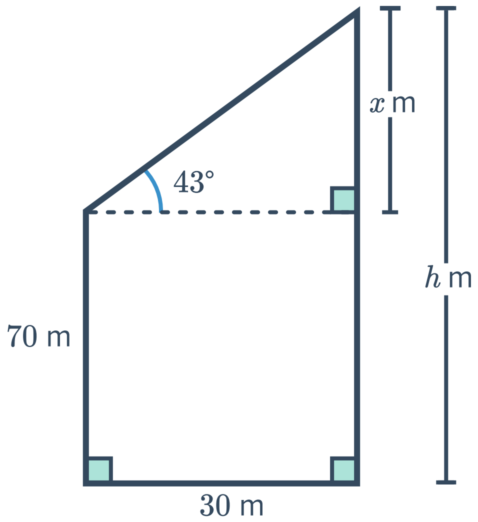
a



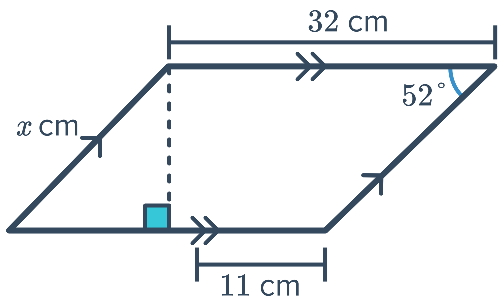
b



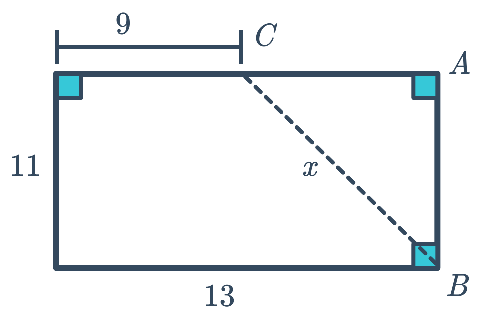
c



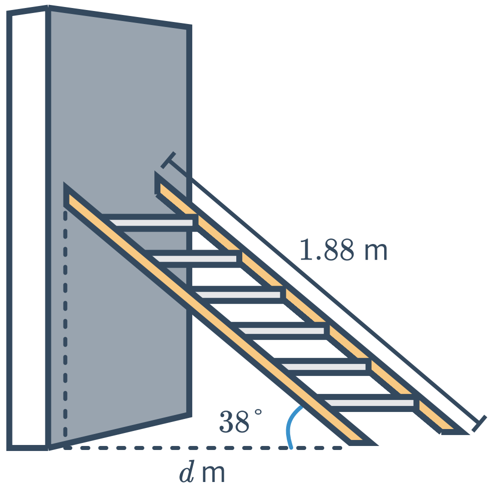
d



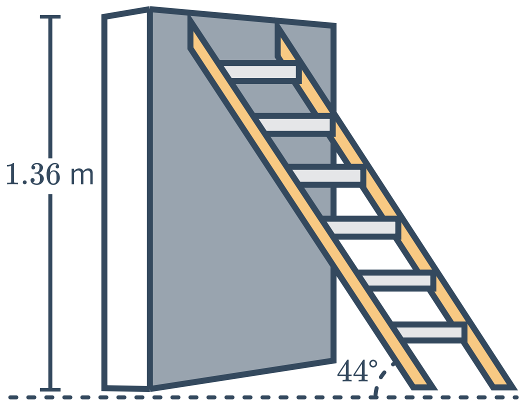
e



16 If d is the distance between the base of the wall and the base of the ladder, find the value of d to two decimal places.

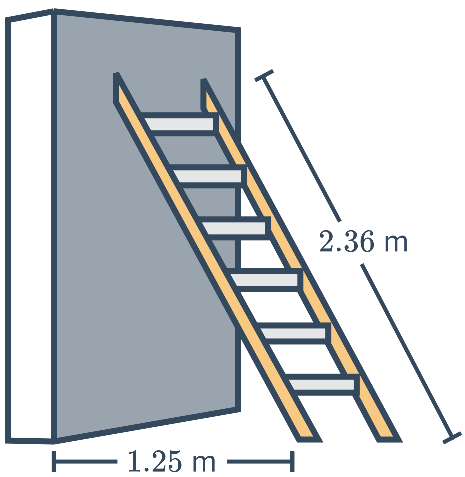


17 A ladder is leaning at an angle of 44° against a 1.36 m high wall. Find the length of the ladder, to two decimal places.



18 A ladder measuring 2.36 m in length is leaning against a wall.

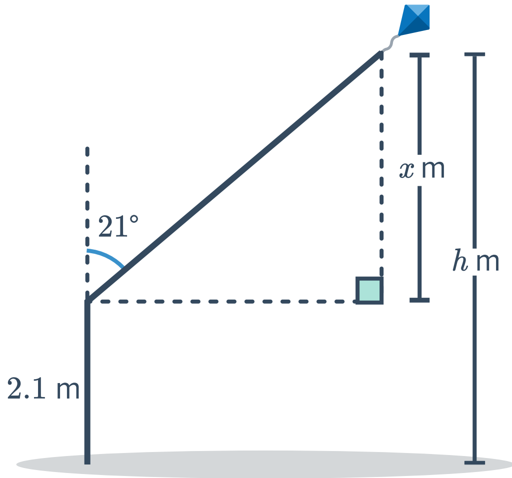
If the angle the ladder makes with the ground is y°, find the value of y to two decimal places.



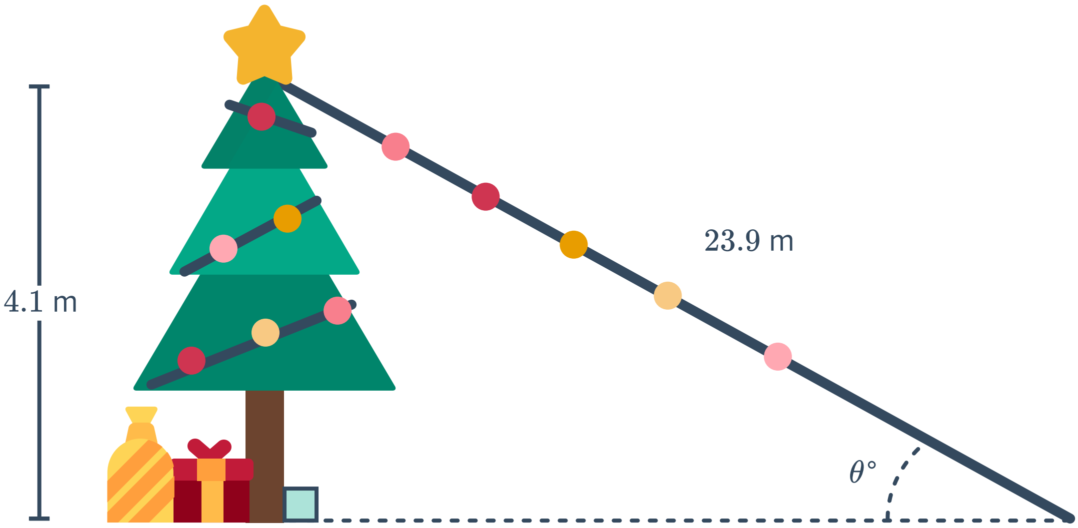
19 A girl is flying a kite that is attached to the end of a 23.4 m length of string. The angle between the string and the vertical is 21°. The girl is holding the string 2.1 m above the ground.

a Find x, correct to two decimal places.

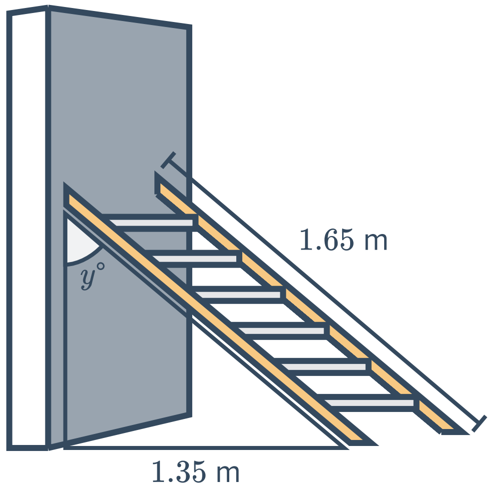
b Hence, find the height, h, of the kite above the ground, correct to two decimal places.



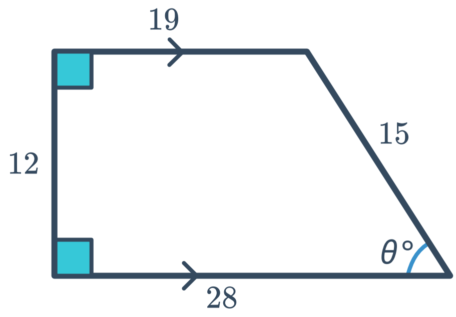
20 In the diagram, a string of lights joins the top of the tree to a point on the ground 23.9 m away. If the angle that the string of lights makes with the ground is θ°, find θ to two decimal places.



21 A ladder measuring $1.65 \mathrm{~m}$ in length is leaning against a wall. If the angle the ladder makes with the wall is y°, find y to two decimal places.



22 Find the value of $\tan \theta$ in the following trapezium:



23 A sand pile has an angle of 40° and is $10.6 \mathrm{~m}$ wide.

Find the height of the sand pile, h, to one decimal place.